Cairo University

Faculty of Engineering

Electronics and Electrical Communications Engineering

Assignment 3

ELC 4028 – Artificial Neural Networks

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1 Problem 1 VAE Synthetic Data with Low-Data Stabilization

1.1 Conditional Variational Autoencoder (VAE)

In this experiment, a Conditional Variational Autoencoder (CVAE) was implemented to generate synthetic MNIST handwritten digit images using only a limited real dataset of 350 samples per digit.

The main objective was to investigate whether synthetic data generated by a VAE can improve the performance of a LeNet-5 classifier trained on a small dataset.

1.1.1 Pipeline Overview

The implemented pipeline is summarized as follows:

1. Load and preprocess MNIST dataset.
2. Reduce the training dataset to only 350 real samples per digit.
3. Apply data augmentation using:
   - Rotation
   - Translation
   - Scaling
   - Gaussian noise
4. Train a Conditional VAE using the augmented dataset.
5. Repeat VAE training for 5 independent runs using different random weight initialization.
6. Generate 1000 synthetic images per digit from each VAE run.
7. Train a LeNet-5 classifier using only the 350 real samples.
8. Classify generated samples and compute:

$$\text{confidence} = \max(\text{softmax output})$$

9. Create three synthetic datasets:
   - Set A: all generated samples
   - Set B: confidence ≥ 0.9
   - Set C: $0.6 \leq$ confidence < 0.9
10. Retrain LeNet-5 using:
    - 350 real samples only
    - 350 real + Set A
    - 350 real + Set B
    - 350 real + Set C

11. Compare against:
    - 350-real baseline
    - 1000-real baseline

1.1.2 Pipeline Flowchart

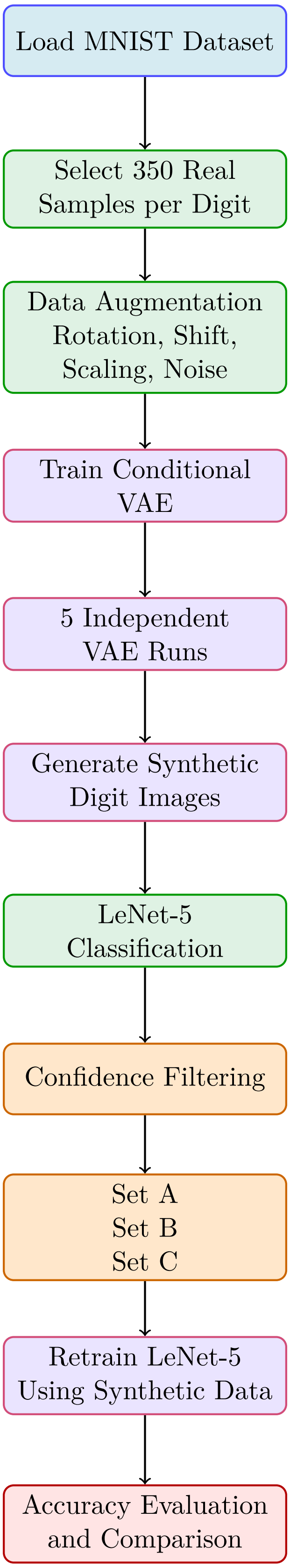

Figure 1.1: Colored pipeline of the Conditional VAE synthetic data generation framework.

1.1.3 Original MNIST Samples

Figure 1.2 shows examples from the original MNIST dataset.

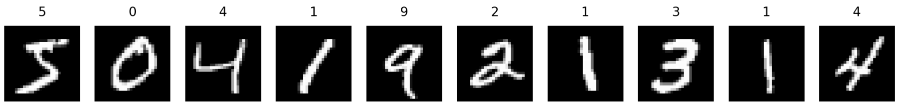

Figure 1.2: Original MNIST samples.

1.1.4 Data Augmentation

To compensate for the limited dataset size, multiple augmentation techniques were applied.

These augmentations improve generalization and help the VAE learn digit variations. Figure 1.3 shows augmented examples.

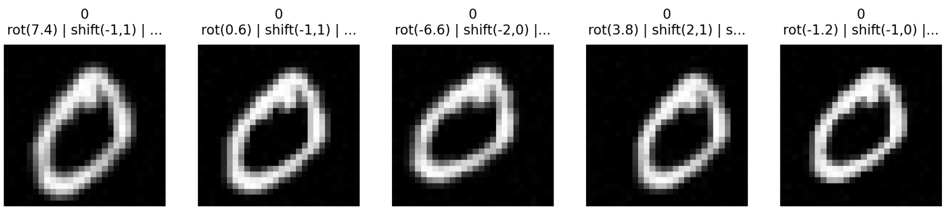

Figure 1.3: Examples of augmented MNIST samples using rotation, shifting, scaling, and noise injection.

1.1.5 VAE Architecture

The implemented Conditional VAE consists of:

- Encoder:
  - Convolutional layers
  - Latent mean and variance estimation
- Decoder:
  - Dense projection
  - Reshape layer
  - Transposed convolution layers
  - Sigmoid output layer

The latent dimension was selected as:

$$z \in \mathbb{R}^8$$

The loss function combines:

- Binary Cross Entropy reconstruction loss
- KL-divergence regularization

1.1.6 Generated Samples During Training

Five independent VAE runs were trained using different random initialization.
Figures 1.4 – 1.8 show generated samples from different runs.

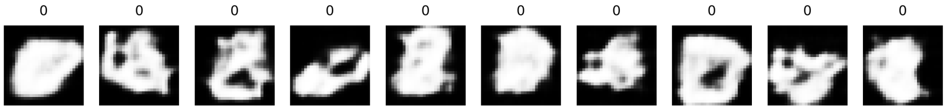

Figure 1.4: Generated samples from VAE Run 1.

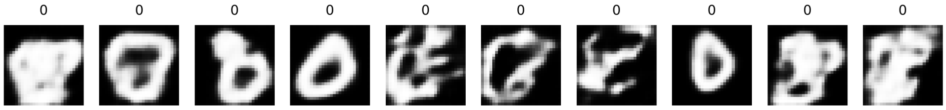

Figure 1.5: Generated samples from VAE Run 2.

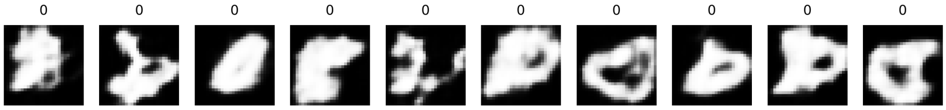

Figure 1.6: Generated samples from VAE Run 3.

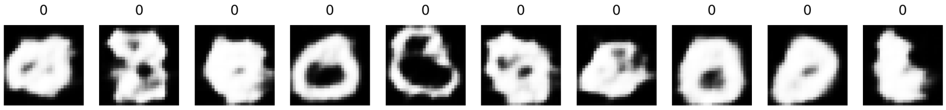

Figure 1.7: Generated samples from VAE Run 4.

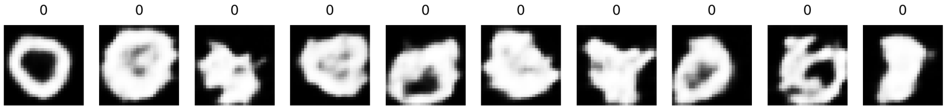

Figure 1.8: Generated samples from VAE Run 5.

The generated images demonstrate that the Conditional VAE successfully learned the structural characteristics of handwritten digits while maintaining variation across samples.

1.1.7 Synthetic Dataset Construction

Generated samples were filtered using the trained LeNet-5 classifier confidence.

Three datasets were created:

- **Set A**: all generated samples
- **Set B**: high-confidence samples
- **Set C**: mid-confidence samples

1.1.8 Set A Samples

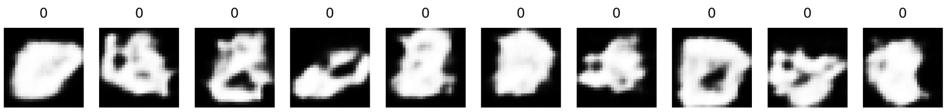

Figure 1.9: Samples from Set A (all generated samples).

Set A contains all generated images without filtering. Some samples are realistic while others contain distortions and noisy structures.

1.1.9 Set B Samples

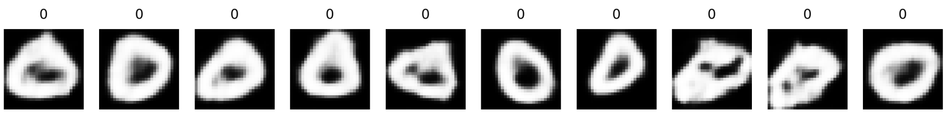

Figure 1.10: Samples from Set B (high-confidence samples).

Set B contains cleaner and more realistic samples because only high-confidence predictions were accepted.

1.1.10 Set C Samples

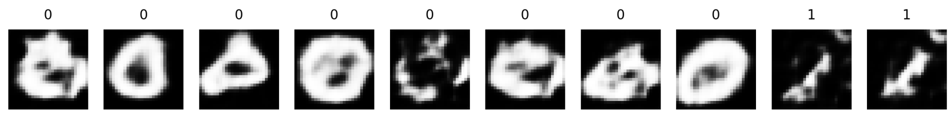

Figure 1.11: Samples from Set C (mid-confidence samples).

Set C contains moderately confident samples that preserve diversity while removing highly corrupted outputs.

1.1.11 Experimental Results

The final LeNet-5 classification accuracies are shown in Table 1.1.

Table 1.1: Comparison of classifier accuracy using different synthetic datasets.

Experiment	**Accuracy**
350 Real Baseline	96.36%
Set A	93.31%
Set B	95.25%
Set C	95.29%
1000 Real Baseline	98.17%

1.1.12 Discussion

The experimental results show that:

- Using only 350 real samples achieved a strong baseline accuracy of 96.36%.
- Adding all generated samples (Set A) reduced performance due to the inclusion of low-quality and distorted generated images.
- Confidence filtering significantly improved performance.
- Set B and Set C achieved accuracies close to the real-only baseline.
- High-confidence filtering successfully removed unrealistic samples while preserving useful digit variations.
- The 1000-real baseline still achieved the best performance because real samples preserve the true MNIST distribution more accurately than synthetic samples.

1.1.13 Conclusion

A Conditional Variational Autoencoder was successfully implemented for MNIST handwritten digit generation.

The generated samples demonstrated that the VAE learned meaningful latent representations despite training on a very limited dataset.

Experimental evaluation showed that synthetic data quality strongly affects classifier performance. Confidence-based filtering improved the usefulness of generated samples and produced classification accuracy close to the real-only baseline.

The results indicate that VAE-generated data can partially compensate for limited datasets, although synthetic samples still do not fully replace real training data.

2 Problem 2 CNN Experiments (LeNet-style)

2.1 Place for Problem 2

We test

3 Problem 3 Speech Recognition from Speech Spectrum with Augmentation

3.1 Place holder for problem 3

test

4 Problem 4 Utterance Representations and SVM Classification

4.1 Place holder for problem 4

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5 Problem 5 Data Augmentation

5.1 Place holder for Problem 1

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